VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × GAS SAFETY

PORTABLE GAS DETECTION

Portable Single Gas Detector *Pocket-sized*

One gas, one sensor, one job — a palm-sized detector that warns the person wearing it before a toxic, flammable or oxygen-deficient atmosphere becomes a casualty.



< 30 s
RESPONSE TIME T90

15 hr
CONTINUOUS BATTERY RUN

< 200 g
CLIP-ON WEIGHT

4 SENSOR TYPES | **2-LEVEL** LOW & HIGH ALARM | **FIELD** ZERO & SPAN

0.1
DISPLAY RESOLUTION

±0.2%
FS
MEASUREMENT ACCURACY

<30s
RESPONSE TIME T90

3–4yr
FIELD-REPLACEABLE SENSOR LIFE

01 OVERVIEW

A sensor that goes *where the worker goes.*

A single-gas detector is the simplest layer of a gas safety system: a compact monitor dedicated to one target gas, worn or carried into the area where that gas could be present. It alarms locally so the person can leave, and it can be read out to a control system so a process can be shut down.

Built around an embedded MCU with a graphical display, two user-set alarm thresholds and a plug-in sensor head — the housing, display and alarms stay identical whichever sensor is fitted.

SENSOR FAMILIES

- ELECTROCHEMICAL
- CATALYTIC BEAD
- NDIR
- PID
- DIFFUSION
- PUMPED OPTION
- LOW / HIGH ALARM
- USB OUTPUT

· PRODUCT FEATURES

Two-Level Alarm

User-presettable Low and High thresholds trigger a loud buzzer, bright LEDs and vibration — noticeable through gloves and hearing protection.

Swap-In Sensor Head

Sensors last three to four years and are replaced by the user in minutes — no return to a service centre, no downtime.

15-Hour Shift Battery

A full charge covers a long shift with margin. Charging is through the same USB port used for data download.

Field Calibration

Zero and span calibration from the keypad with standard calibration gas, guided remotely by Vepolink engineers when needed.

02 SENSING PRINCIPLES

Four ways to *see a gas.*

No single sensor covers every hazard. The detector accepts four sensor technologies, each matched to the chemistry of the gas it is asked to find.

ELECTROCHEMICAL

Toxic Gases & Oxygen

CO · H_2S · SO_2 · NH_3 · CL_2 · O_2



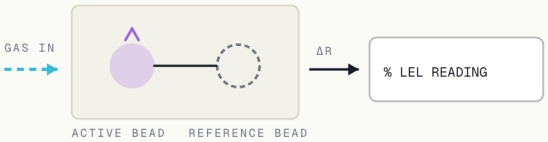
Gas diffusing into the cell is oxidised or reduced at the sensing electrode, producing a current in direct proportion to concentration. Low power, ppm resolution, and selective by electrode chemistry.

`Current ∝ concentration`

CATALYTIC BEAD

Combustible Gases

CH_4 · LPG · H_2 · HEXANE · ETHYLENE



Flammable gas burns on a heated catalytic bead, raising its resistance relative to an inert reference bead. The imbalance reads out as percent of lower explosive limit.

`Reads in % LEL`

NDIR

Carbon Dioxide & Hydrocarbons

NON-DISPERSIVE INFRARED ABSORPTION



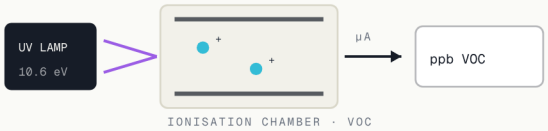
An infrared beam crosses a gas cell at a wavelength the target absorbs. The drop in light reaching the detector gives concentration — with no oxygen needed and no poisoning of the sensor.

`Absorption ∝ concentration`

PID

Volatile Organic Compounds

10.6 EV PHOTO-IONISATION



Ultraviolet light ionises organic vapours passing between two electrodes; the resulting ion current is measured. Sensitive to the ppb level for solvents, fuels and many industrial VOCs.

`VOC screening`

03 GAS COVERAGE

Pick the gas. *Fit the sensor.*

Each unit is supplied configured for one target gas. The most requested configurations are listed below; further gases are available on request.

O_2	Oxygen	EC	CO	Carbon Monoxide	EC	H_2S	Hydrogen Sulphide	EC	SO_2	Sulphur Dioxide	EC
NH_3	Ammonia	EC	Cl_2	Chlorine	EC	HCl	Hydrogen Chloride	EC	HCN	Hydrogen Cyanide	EC
NO	Nitric Oxide	EC	NO_2	Nitrogen Dioxide	EC	PH_3	Phosphine	EC	$COCl_2$	Phosgene	EC
O_3	Ozone	EC	H_2	Hydrogen	CAT	LPG	LPG / Propane	CAT	CH_4	Methane · Combustibles	CAT
C_6H_{14}	Hexane	CAT	C_2H_4	Ethylene	CAT	CO_2	Carbon Dioxide	NDIR	VOC	Volatile Organics	PID

STEP 01

Bump and go

Power on, watch the self-test, confirm the sensor responds to a short burst of test gas, then clip it on.

STEP 02

Sample

Gas reaches the sensor by diffusion through the grille, or is drawn in by the optional auto-suction pump for confined-space pre-entry checks.

STEP 03

Alarm

Crossing the Low threshold warns; crossing High demands evacuation. Buzzer, LEDs and vibration fire together.

STEP 04

Log and review

Readings and alarm events are stored on board and downloaded over USB for exposure records and incident review.

DIFFUSION OR PUMPED

In diffusion mode the sensor breathes the air around the wearer — the right choice for personal protection. The auto-suction option adds a small pump and probe so the atmosphere in a tank, pit or manhole can be tested from outside before anyone enters.

SETTING ALARM LEVELS

Thresholds are set from the keypad and typically follow occupational exposure limits — for example a Low alarm at the TWA value and High at the STEL, or 19.5 % and 23.5 % for oxygen. Values are retained through power cycles and sensor changes.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

Sensor Technology	Electrochemical · Catalytic · NDIR · PID
Target Gases	Combustibles (CH_4, C_3H_8, H_2), oxygen, toxic gases incl. CO, H_2S, SO_2, NH_3, NO, NO_2, HCN, PH_3, Cl_2
Resolution	0.1 (unit per gas: ppm, % vol or % LEL)
Accuracy	±0.2 % of full scale
Response Time	T90 < 30 s
Sampling Type	Diffusion · auto-suction pump optional
Sensor Life	3 - 4 years, user replaceable
Zero Calibration	Yes, from keypad
Span Calibration	Yes, with standard calibration gas

ALARM · POWER · BUILD

Display	Digital graphical display, backlit
Controller	Embedded MCU
Alarm Levels	2 level · Low & High, user presettable
Alarm Indication	Buzzer · LED · vibration
Battery Life	Up to 15 hours continuous operation
Charging	230 VAC, 50 Hz adaptor via USB
Digital Output	USB
Weight	Less than 200 g
Housing	Impact-resistant, rubberised, with belt clip

INSTRUMENT DIMENSIONS

INDICATIVE FRONT & SIDE ELEVATION · MM



WEIGHT
< 200 g

05 USE & UPKEEP

Clipped on, checked, *trusted.*

> A personal monitor is only as good as its last check. Good practice is a **bump test before every shift**, a **full calibration at scheduled intervals** with certified gas, and a sensor change when response slows — all of which the user can do without sending the unit away.

01 · WEARING

In the breathing zone

Clip to a collar, lapel or chest pocket so the sensor samples the air the wearer breathes, grille unobstructed.

02 · BUMP TEST

Daily, seconds

A short exposure to test gas confirms the sensor and alarms respond before the unit goes into a hazardous area.

03 · CALIBRATION

Zero and span

Zero in clean air, span with a known concentration. Remote guidance from Vepolink engineers is available for the first few cycles.

04 · CHARGING

Overnight over USB

Return to the charger at shift end. A full charge delivers up to 15 hours, with a low-battery warning well before shutdown.

05 · SENSOR CHANGE

User serviceable

Sensors last three to four years and swap out in minutes. Recalibrate after fitting and the unit is back in service.

06 · RECORDS

Downloaded over USB

Logged readings and alarm events export to a PC for exposure history, audits and incident investigation.

SINGLE GAS OR MULTI GAS?

Where the hazard is known and singular — H_2S at a sour-gas facility, CO in a boiler house, oxygen in a nitrogen-blanketed area — a single-gas unit is lighter, cheaper and simpler to maintain than a multi-gas instrument. Where the atmosphere is uncertain, the Vepolink portable multi-gas detector monitors four gases in one housing and is the better fit for confined-space entry.

FLEET SUPPLY

Units are supplied individually or as a site fleet with calibration gas, regulators, charging cradles and a spares kit of sensors and filters, so the safety team can keep every detector in service from its own store.

06 APPLICATIONS

Where it *works.*

Anywhere a single, known gas is the hazard — and anyone whose job puts them in its path.

Refineries & Petrochemicals	Chemical & Fertiliser Plants	Confined-Space Entry
Sewers & Water Treatment	Power & Boiler Houses	Steel & Metallurgy
Pharma & Laboratories	Cold Storage & Ammonia Plants	Mining, Tunnels & Shipyards

PERSONAL AND PROCESS PROTECTION

The detector serves two audiences: the individual, who gets an immediate warning to leave, and the site, which gets a logged record of every exposure and alarm event. Paired with fixed detectors and a control interface, a portable unit closes the gap in the places fixed heads cannot reach — inside vessels, under floors, at the end of a hose.

CHOOSING THE GAS

The right configuration follows the process: H_2S for sour crude and sewage, CO for combustion and steel, NH_3 for refrigeration, Cl_2 for water treatment, oxygen wherever inert gas or displacement is a risk. Tell us the process and we will recommend the sensor and alarm levels.

07 FAQ

Frequently asked *questions.*

Q What is the sampling method?

Both diffusion and auto-suction types are offered. Diffusion suits personal wear; the pumped version draws a sample through a probe for pre-entry checks.

Q Can the user replace the sensor?

Yes. The sensor head is a plug-in module that can be replaced by the user at any time, followed by a recalibration.

Q Can the user calibrate the unit?

Yes. With standard calibration gas, zero and span calibration is done from the keypad — with remote help from Vepolink engineers if wanted.

Q Which gases can be detected?

Oxygen, combustibles, LPG, hydrogen, CO, H_2S, Cl_2, SO_2, NH_3, HCl, HCN, NO, NO_2, ethylene, phosphine, phosgene, ozone, CO_2, VOCs and more — one gas per unit.

Q How long does a sensor last?

Typically three to four years in normal service. Response slows as a cell ages, which shows up at bump test or calibration.

Q How are alarms set?

Two levels — Low and High — are user presettable from the keypad and retained in memory. All three indicators (buzzer, LED, vibration) fire together.

Q How long does the battery last?

Up to 15 hours of continuous operation on a full charge — a long shift with margin. Charging is over USB from a 230 VAC adaptor.

Q Can readings be downloaded?

Yes. Logged readings and alarm events are exported over the USB port for exposure records and incident review.

Q Can it interface with a control system?

Portable units alarm locally. For automatic process shutdown, pair them with Vepolink fixed gas detectors wired to the plant control system.

Q How heavy is it?

Under 200 g with clip — light enough to be worn all shift without noticing it.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. Gas detectors are safety equipment: follow the user manual, site procedures and applicable standards for bump testing, calibration and use. Specifications may change without notice as the product is improved.

VEPOLINK
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ONE GAS, WATCHED ALL SHIFT

Protect the person. *Not just the plant.*

Tell us the gas, the process and how many people need protecting, and we'll configure the detectors, alarm levels, calibration gas and charging kit as one supply. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specifies, calibrates and supports gas detection. No call centre.
+91 99538 86178 →

02 **Request a fleet quotation** RECOMMENDED
Send your gas list and headcount — we'll spec sensors, alarm levels, calibration gas, chargers and spares.
sales@vepolink.com →

03 **Book a demonstration**
See a bump test, a calibration and a sensor change with our engineering team.
Request a demo →

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